

MALAINA CHOY

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TECHNICAL SKILLS

Research & Engineering: Experimental design, verification & validation testing, data collection & analysis, prototyping, 3D printing, sensor integration, technical documentation

Programming & Software: Python, MATLAB, ROS2, GitHub, SolidWorks, Linux, Microsoft Office

Certifications: CSWA – Mechanical Design, CSWA-S – Simulation Associate

EDUCATION

The University of Western Ontario, London, ON | Expected Graduation: May 2027

Bachelor of Engineering Science, Mechanical Engineering and concurrent Biomedical Engineering

- **Relevant Coursework:** Engineering Design, Control Systems, Finite Element Analysis, Medical & Assistive Devices, Biomechanics, Biomaterials, Biomedical Engineering Design
 - Dean's Honour List
 - Andrea Bailey Memorial Award 2024, 2025
 - Lynda Diane Shaw Memorial Award 2025
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RESEARCH EXPERIENCE

Undergraduate Research Student, Western Engineering Spine and Trauma Lab | *October 2025-Present*

- Develop and validate an ultrasound-based system for 3D reconstruction of bone, integrating medical imaging, motion tracking, sensor data, and robotic systems
- Design and execute repeatable experimental testing to evaluate accuracy and system performance against robotic ground-truth measurements, identifying sources of error and supporting iterative system improvements
- Integrate and troubleshoot hardware and software subsystems, investigating tracking, synchronization, sensor, and data-quality issues during experimental operation
- Develop Python/ROS2 workflows for data acquisition, logging, processing, and quantitative analysis
- Document experimental methods, validation procedures, and technical results, and communicate findings to a multidisciplinary research team

Undergraduate Summer Research Student, Knee Biomechanics Lab | *May 2025-August 2025*

- Designed and prototyped a wearable biomechanical sensing system integrating triboelectric sensors and load cells to measure plantar loading during movement
- Fabricated and experimentally tested sensor-integrated prototypes under varying loading conditions to evaluate device performance and repeatability
- Collected and analyzed quantitative biomechanical data to characterize sensor response to applied loads and support iterative prototype evaluation
- Presented findings at Western University's Undergraduate Summer Research Symposium

Additional Experience

Private Tutor — *August 2025 - Present*

Senior Math & Science Tutor — Oxford Learning | *May–Aug. 2024*

LEADERSHIP & INVOLVEMENT

Canadian Undergraduate Biomedical Engineering Council, Marketing Lead of Ontario | *May 2026-Present*

True North Biomedical Competition, Communications Commissioner | *June 2025-Present*

Alzheimers Society Southwest Partners, In-home Recreation Volunteer | *June 2025-Present*

Humane Society London and Middlesex, Front-desk Support & Animal Care Partner | *February 2026-Present*

Western Engineers Without Borders, Structural Build Sub-team Lead | *July 2025-May 2026*